

Karim Zaghw

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Research Profile

MSc student in Computational Neuroscience at the University of Tübingen, working primarily on machine learning. I study how learning agents can quantify epistemic uncertainty and form causal or hierarchical representations that support planning and generalization. Current work spans distributional reinforcement learning, Bayesian planning, causal representation learning, and hierarchical generative models.

Education

MSc Computational Neuroscience

University of Tübingen
Expected graduation: 03/2027.

2024/10–Present
Tübingen, Germany

BSc Computer Engineering

Cairo University
GPA: 3.74/4.00.

2016/09–2021/05
Cairo, Egypt

Publications

Karim Zaghw, Peter Dayan, and Georgy Antonov (2026). *Weight a Moment: Risk-Aware Exploration with Bayes*. Accepted, RLC 2026 / RLJ; camera-ready completed; [OpenReview](#).

Karim Zaghw, Andrew Pashea, Marc Pritsch, Wouter Nuijten, Karl Friston, and Lancelot Da Costa (2026). “Renormalising Generative Models for Active Inference: Foundations, Derivations, and Verification”. Accepted, IWAI 2026; preprint forthcoming.

H. Osman, **Karim Zaghw**, M. Hazem, and S. Elsehely (2020). “An Efficient Language-Independent Multi-Font OCR for Arabic Script”. In: *The International Conference of Digital Image Processing and Pattern Recognition (DPPR)*. [arXiv](#); [presentation](#).

Research Experience

Student Research Assistant

Department of Computational Neuroscience, Max Planck Institute for Biological Cybernetics

2025/01–Present
Tübingen, Germany

- Developed Bayesian Moment Learning, a model-free distributional reinforcement-learning method that represents epistemic uncertainty over risk-relevant moments and extends value-of-perfect-information exploration to moment-based risk objectives; work accepted at RLC 2026 / RLJ.
- Current project: developing Bayesian Monte Carlo tree search with Normal–Gamma edge posteriors, Bayesian Q-learning-style backups, and posterior-probability and value-of-computation stopping criteria.

Lab Rotations

Lab Rotation, Causal Abstraction for Overcomplete Representations

Robust Machine Learning Group, ELLIS Institute Tübingen

2026/07/01–2026/09/23
Tübingen, Germany

- Studying when block-identifiable overcomplete representations preserve high-level causal abstractions under reparameterization.

Lab Rotation, Renormalising Generative Models

Methods of Machine Learning Group, University of Tübingen

2026/04/15–2026/06/16
Tübingen, Germany

- Reconstructed and verified a hierarchical active-inference framework, translating core MATLAB/SPM routines into Python and clarifying its inference, planning, and structure-learning derivations; work accepted at IWAI 2026.

Teaching and Mentorship

Internship Mentor, [Raisa Energy](#)

Mentored a Text-to-SQL chatbot project using LLMs; achieved over 90% accuracy on daily queries; [project write-up](#).

2023/07–2023/10

Junior Teaching Assistant, Operating Systems, Cairo University

Assisted in teaching, writing, and grading assignments, projects, and quizzes for 100+ students.

2021/01–2021/06

Additional Training

[MITx MicroMasters Program in Statistics and Data Science](#)

MITx (MIT Open Learning), edX; SDS General Track.

2023/02–2024/04

[Deep Learning Specialization](#), [DeepLearning.AI](#)

2022/12–2023/05

Professional Experience

Data Scientist / Data Engineer, [Raisa Energy](#)

Built probabilistic forecasting, risk-modeling, and data-engineering systems used across 20+ projects and \$1B+ of tracked assets.

2021/07–2024/09